NAISC

Adaptive Drift Monitoring System Report

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1. Executive Summary

This project presents an end-to-end **Adaptive Data Drift Monitoring and Mitigation System** for telecom churn prediction. The system is designed to detect, quantify, and correct data drift occurring between training and production environments, ensuring consistent model performance under non-stationary conditions.

At a high level, the pipeline integrates drift detection, statistical validation, mitigation strategies, and model retraining using a LightGBM classifier. Performance is evaluated using AU-PRC, comparing baseline and drift-mitigated models.

System Outcome Summary

Component	Output
Drift Detection	Feature-level PSI + statistical tests
Mitigation	Encoding + scaling + sample reweighting
Model	LightGBM (baseline + mitigated)
Evaluation	AU-PRC comparison
Artifacts	Drift table, predictions, reports

The system demonstrates improved robustness after mitigation, validating the effectiveness of drift-aware learning.

2. Problem Definition

In production machine learning systems, data is not static. In telecom churn prediction, customer behavior changes over time due to pricing updates, service upgrades, competitive pressure, and evolving usage patterns. This leads to performance degradation when models trained on historical data are deployed in live environments.

This issue is known as **data drift**, where the statistical properties of input features change even though the underlying target concept (customer churn behavior) remains stable.

Key Challenges

- **Feature distribution shift**
  Changes in input feature distributions reduce the relevance of learned patterns and degrade model performance.
- **Unseen categories in production**
  New categorical values can appear in live data, leading to encoding issues or biased representations.

- **Numerical distribution changes**
  Shifts in scale, variance, or skewness of continuous features reduce model generalization ability.
- **Class imbalance effects**
  Imbalanced churn labels can distort evaluation metrics and make drift-related degradation harder to detect.

The objective of this system is to build a **drift-aware machine learning pipeline** that can detect, quantify, and mitigate data drift automatically.

This is achieved through:

- Monitoring feature distribution changes over time
- Identifying drift severity across features
- Applying mitigation strategies such as reweighting and retraining

The final goal is to ensure **stable and reliable model performance in dynamic production environments**.

3. System Overview

The system follows a modular, production-oriented pipeline designed for clarity, scalability, and separation of concerns.

Pipeline Flow

Design Principle

The system is intentionally structured as a **linear but modular pipeline**, where each stage produces a well-defined output that becomes the input to the next stage.

4. Data Understanding

The dataset consists of structured telecom customer records designed for churn prediction. It captures customer demographics, service usage behavior, financial patterns, and contract-level attributes. A detailed exploratory data analysis (EDA) was performed to understand data quality, distributional properties, and potential modeling risks.

4.1 Dataset Structure

The dataset contains **70,430 customer records** with **45 features**, including temporal, categorical, and numerical attributes.

Feature Type	Count	Examples
Categorical	28	Contract type, Payment method, Internet type
Numerical	17	Charges, usage metrics, tenure
Temporal	1	Month
Target	1	ChurnStatus (binary classification)

A key observation from EDA is that the dataset is **feature-rich but distributionally heterogeneous**, meaning different feature groups behave under different statistical patterns.

4.2 Missing Values & Semantic Interpretation

A significant portion of missing values were identified in service-related features:

Feature	Missing Volume	Interpretation
Offer	~48k	No promotional offer applied
InternetType	~17k	Customer not subscribed to internet service
PrioritySupport	~12k	No premium support plan
DigitalInvoicing	~12k	Customer uses paper billing

Rather than representing data corruption, these missing values are **informative nulls**, encoding the absence of a service. This required careful handling during preprocessing by treating them as explicit categories instead of imputing them blindly.

4.3 Distributional Insights

The dataset shows realistic telecom customer behavior patterns across demographic, geographic, and financial dimensions.

Demographics & Tenure

- Customer age is concentrated between **45–50 years**, forming a near bell-shaped distribution.
- Tenure exhibits a **bimodal pattern**, with clusters of:
    - New customers (1–12 months)

    - Long-term customers (30–72 months)
- Dependents feature is highly sparse, indicating most customers are individual subscribers.

Geographic Structure
Although restricted to California, spatial analysis reveals two distinct clusters:

- Northern California customer concentration
- Southern California customer concentration
  This suggests regional segmentation effects in service usage patterns.

Financial Features
Revenue-related variables such as `TotalCharges`, `TotalRevenue`, and `TotalLongDistanceCharges` show **strong right skewness**, indicating the presence of a small number of high-value customers.

This motivates:

- Log transformation for stability
- Robust scaling to reduce outlier sensitivity

4.4 Target Distribution & Modeling Implications

The target variable `ChurnStatus` is moderately imbalanced:

- **No Churn:** ~71%
- **Churn:** ~29%

This imbalance is significant enough to impact model learning, especially for minority-class recall. As a result, the modeling strategy incorporates class-aware training using LightGBM's imbalance handling (`is_unbalance = True`) and drift-aware reweighting during mitigation.

4.5 Temporal Analysis (Drift Monitoring Baseline)

The `Month` feature enables simulation of temporal stability checks within the training data.

Key observations:

- Contract types remain stable across time (no major distribution shift)
- Internet service mix is consistent, with Fiber Optic dominating usage
- Churn rate remains relatively stable over observed months

This indicates that the training dataset is largely **stationary (IID-like)** internally, providing a stable baseline before introducing production drift scenarios.

4.6 Drift Behavior Analysis (Experimental Validation)

Although no strong natural drift is observed within the training set, controlled experiments were conducted to evaluate system sensitivity.

Numerical Drift Findings

- Most features show negligible drift under normal conditions (PSI < 0.001)
- Controlled shifts successfully triggered detectable drift:
  - `UserAge`: severe drift (PSI ~0.46–2.57, KS p-value = 0.0)
  - `TotalRevenue`: strong drift (PSI ~0.70)
  - `MonthlyCharge`: moderate drift (PSI ~0.12)
  - `DataUsageAvg`: detectable statistical shift despite low PSI (~0.03)

Key Insight:
Different statistical methods capture different drift behaviors:

- PSI → distribution shift magnitude
- KS Test → sensitivity to shape changes
- Wasserstein Distance → mean/scale movement

This validates the need for a **multi-metric drift detection framework**, rather than relying on a single statistical test.

4.7 Key Conclusion from EDA

Overall, the dataset is:

- Structurally rich and well-formed
- Internally stable (no strong natural drift)
- Statistically suitable for supervised learning

However, it contains realistic complexities such as:

- Informative missingness
- Skewed financial distributions
- Class imbalance
- Latent geographic segmentation

These characteristics justify the use of a **drift-aware modeling pipeline**, where robustness is not assumed but explicitly engineered through detection, scoring, and mitigation stages.

5. Drift Detection Framework

Drift detection is implemented in `drift_detector.py` and operates at a **feature-level granularity**, separately handling categorical and numerical variables. The framework combines statistical distance measures and hypothesis testing to ensure reliable detection of distribution shifts in production-like conditions.

Rather than relying on a single metric, the system uses a **multi-signal detection strategy**, improving robustness across different types of drift behavior.

5.1 Methods Used

Different statistical methods are applied depending on the feature type:

Feature Type	Methods Used
Categorical	PSI, Chi-square test, missingness handling
Numerical	PSI, KS test, Wasserstein distance

This hybrid approach ensures that both:

- **distributional magnitude shifts** (PSI)
- **shape and statistical differences** (KS test, Chi-square)
- **distance-based changes in value distribution** (Wasserstein)

are effectively captured.

5.2 Feature-Level Drift Output

For every feature, the system generates a structured drift profile containing multiple indicators.

Metric	Description
PSI Score	Measures magnitude of distribution shift between train and production data
p-value	Statistical significance from hypothesis testing
Drift Flag	Binary indicator of whether drift is detected
Severity	Interpretable classification of drift strength

This allows the system to support both **quantitative evaluation** and **decision-level interpretation** for downstream mitigation.

5.3 Severity Classification

Drift severity is primarily determined using the PSI score, which provides an interpretable measure of distribution shift intensity.

PSI Range	Severity Level
< 0.10	Low / No Drift
0.10 – 0.25	Moderate Drift
≥ 0.25	High Drift

This rule is directly implemented in `classify_psi()` and ensures consistent interpretation across all features.

5.4 Drift Decision Logic

A feature is classified as drifted based on a **dual-condition rule**, ensuring both statistical and practical significance are considered.

A feature is flagged as drifted if:

- PSI ≥ 0.10, **or**
- p-value < 0.05 (statistical significance)

This design avoids over-reliance on a single metric and reduces false positives caused by noisy distributions.

5.5 System Output Structure

The final output is a ranked drift table sorted by PSI score, along with a list of detected drifted features.

The drift table typically includes:

- Feature name
- Type (Categorical / Numerical)
- PSI score
- Severity label
- Statistical test outputs (KS / Chi-square / Wasserstein)
- Drift flag

This structure enables:

- Feature prioritization for mitigation
- Clear interpretability for model debugging
- Direct integration into the mitigation pipeline

The framework intentionally combines **distribution-based (PSI)**, **statistical hypothesis testing (KS / Chi-square)**, and **distance-based (Wasserstein)** measures to ensure drift is captured from multiple perspectives.

This makes the system more robust in real-world production settings where drift can be

6. Drift Scoring System

The drift scoring system aggregates feature-level drift signals into a unified structure that allows both **ranking of high-risk variables** and **system-wide drift assessment**. While the detection module identifies whether drift exists, the scoring layer quantifies and prioritizes its impact across the dataset.

6.1 Core Output

The primary output of this stage is a structured drift report stored as:

Output File	Purpose
`drift_table.csv`	Full feature-level drift metrics including PSI, statistical tests, and severity labels

This file acts as the central artifact for both analysis and downstream mitigation decisions.

6.2 Drift Ranking Logic

Features are ranked primarily using the **PSI score**, which provides an interpretable measure of distributional shift magnitude. Higher PSI values indicate stronger divergence between training and production distributions.

Rank Category	Interpretation
Top 10% PSI	Critical drift sources requiring immediate attention
Mid-range PSI	Moderate instability with potential impact
Low PSI	Stable features with minimal distribution change

This ranking mechanism ensures that mitigation efforts are **prioritized based on impact severity rather than uniform treatment of all features**.

6.3 System-Level Drift Metrics

Beyond individual feature scoring, the system computes aggregated drift statistics to evaluate overall dataset stability.

Key system-level indicators include:

- **Drift Coverage**
  Percentage of features identified as drifted across the entire dataset. This provides a high-level view of how widespread distribution shifts are.
- **Severity Distribution**
  Breakdown of drifted features into Low, Moderate, and High severity categories, enabling quick assessment of whether drift is localized or systemic.

The scoring system transforms raw drift detection outputs into a **prioritized decision framework**, allowing the pipeline to distinguish between:

- isolated feature-level noise, and
- structurally significant distribution shifts that require mitigation or retraining.

This ensures that downstream resources (such as reweighting, transformation, or retraining) are applied efficiently to the most impactful features.

7. Categorical Drift Handling

Categorical features are particularly sensitive to production changes due to the possibility of **new categories, shifting frequencies, and sparse value distributions**. To address this, the system applies a combination of alignment, encoding stability, and drift-aware weighting strategies.

7.1 Techniques Applied

Technique	Purpose
Category alignment	Ensures consistency between train and production feature spaces by handling mismatched categories
Rare category grouping	Reduces sparsity by consolidating infrequent categories into a shared representation
Encoding recalibration	Stabilizes feature representation when category distributions shift
Drift-based weighting	Reduces the influence of unstable or highly drifting categorical values during model training

These techniques are designed to ensure that categorical features remain **consistent, interpretable, and model-ready**, even when production data introduces previously unseen or imbalanced categories.

7.2 Outcome

This handling strategy ensures that categorical inconsistencies do not propagate into downstream modeling stages. In particular, it prevents:

- Encoding failures due to unseen categories
- Bias introduced by rare or unstable categorical values
- Over-sensitivity of the model to distributional noise

Overall, the categorical pipeline improves **robustness and generalization under real-world distribution shifts**.

8. Numerical Drift Handling

Numerical features are more sensitive to **distributional shifts in scale, variance, and outliers**, which can significantly affect model stability. To address this, the system applies statistical correction and normalization techniques that align training and production distributions.

8.1 Techniques Applied

Technique	Function
Distribution normalization	Aligns train and production distributions to reduce statistical divergence
Sample reweighting	Assigns higher or lower importance to samples based on drift intensity
Scaling recalibration	Standardizes feature ranges to maintain consistent magnitude across datasets
Outlier stabilization	Reduces the impact of extreme values that can distort model learning

These methods ensure that numerical features remain **statistically comparable across environments**, even under shifting data distributions.

8.2 Effect

The applied transformations improve model robustness by:

- Reducing sensitivity to production distribution shifts
- Stabilizing feature scales across training and inference
- Limiting the influence of extreme or anomalous values
- Improving generalization under non-stationary conditions

Together, categorical and numerical drift handling form a unified mitigation layer that ensures the model does not rely on static assumptions about feature distributions. Instead, it continuously adapts to distributional changes, improving long-term reliability in production environments.

9. Global Mitigation Strategy

Rather than treating drift as an isolated, feature-by-feature problem, the system applies a **global mitigation framework** that adjusts the learning process holistically. The goal is to ensure that the model adapts to distribution shifts in a controlled manner while preserving predictive stability.

9.1 Strategy Overview

Component	Role
Sample reweighting	Adjusts the importance of training samples based on drift intensity
Feature weighting	Reduces the influence of highly unstable or noisy features
Mixed dataset training	Combines training and production-like data to improve generalization
Drift-aware learning	Integrates PSI-derived signals into the training process to guide model sensitivity

This design ensures that mitigation is not reactive, but instead **embedded into the learning process itself**.

9.2 System Behavior

The global mitigation framework modifies model training dynamics in the following ways:

- Stable features retain their predictive strength and continue to contribute reliably to the model
- Highly drifting features are automatically down-weighted to reduce noise and prevent overfitting to unstable patterns
- The model is exposed to production-like distributions through mixed dataset training, improving adaptability
- PSI-based drift signals guide how strongly each feature influences the final learning process

This approach shifts the system from **static training** to **drift-aware learning**, where the model is explicitly informed about distribution instability during training.

As a result, the model becomes more robust to real-world deployment conditions, maintaining performance even when feature distributions evolve over time.

10. Model Strategy (LightGBM Fixed)

The final predictive model uses **LightGBM**, selected by competition constraints for its efficiency and strong performance on structured tabular data. Importantly, the model architecture and hyperparameters are **fixed and cannot be modified**, making performance improvements dependent entirely on data quality and drift mitigation strategies.

This shifts the focus from model optimization to **data-centric learning and drift-aware training**.

Training Pipeline

Since model parameters are fixed, the system operates using two controlled training setups:

Model Stage	Data Used
Baseline Model	Original training dataset
Mitigated Model	Drift-adjusted dataset with reweighting and preprocessing improvements

Both models use the same LightGBM configuration, ensuring a fair comparison between standard and drift-aware training.

11. Feature Engineering

The feature engineering pipeline was designed to ensure compatibility with the LightGBM model while maintaining robustness under data drift conditions.

Categorical features were preserved in their native form and converted into categorical data types supported by LightGBM. Category alignment was enforced between training and production datasets to prevent unseen category issues during inference. This ensured consistent encoding across environments.

Numerical features were initially used in raw form and later adjusted during the drift mitigation phase using normalization and weighting strategies. No aggressive transformations were applied initially to avoid introducing bias prior to drift detection.

Missing values were handled implicitly by LightGBM, which can naturally manage missing splits. Additionally, feature alignment between the train and test datasets ensured that no feature mismatch occurred during prediction.

This approach prioritised **stability, compatibility, and minimal preprocessing leakage**, ensuring that drift detection and mitigation remained the primary focus of the system.

12. Evaluation Metrics

The primary evaluation metric used in this system is **Area Under the Precision-Recall Curve (AU-PRC)**.

AU-PRC is particularly suitable for telecom churn prediction due to the class imbalance present in the dataset, where churned customers represent a minority class. Unlike accuracy or ROC-AUC, AU-PRC focuses on the model's ability to correctly identify positive cases (churn), which is more critical in business applications.

Performance was evaluated on both:

- Training data (to monitor overfitting)
- Production/test data (to assess real-world generalization)

The comparison between baseline and post-mitigation performance provides a direct measure of the effectiveness of the drift handling strategy.

13. Results Summary

```
============================================================
 MODEL PERFORMANCE COMPARISON
============================================================

Baseline Train AU-PRC  : 0.9255
Mitigated Train AU-PRC : 0.9262
Train Improvement      : +0.0007

Baseline Test AU-PRC   : 0.7516
Mitigated Test AU-PRC  : 0.7677
Test Improvement       : +0.0160
```

The system demonstrated measurable improvements after applying drift mitigation techniques.

Key Results:

- Baseline Train AU-PRC: 0.9255
- Mitigated Train AU-PRC: 0.9262
- Train Improvement: +0.0007
- Baseline Test AU-PRC: 0.7516
- Mitigated Test AU-PRC: 0.7677
- Test Improvement: +0.0160

A total of **14 out of 43 features (32.6%)** were identified as drifted, with **100% monitoring coverage**.

The system successfully improved test performance while maintaining stable training performance, suggesting improved generalization after drift mitigation.

14. Runtime Analysis

The system was designed with efficiency in mind, ensuring suitability for near real-time monitoring scenarios.

Runtime Breakdown:

- Total Drift Analysis Runtime: ~2.07 seconds
- Drift Detection Phase: Majority of runtime
- Model Training (LightGBM): Fast due to optimized implementation
- Mitigation Overhead: Minimal relative to detection

The lightweight nature of the pipeline allows it to scale efficiently across larger datasets and be integrated into production monitoring systems.

15. Drift Insights

Analysis of drifted features reveals several important patterns:

High Impact Drift Features

- Contract
- Month
- TransactionMode
- DigitalInvoicing
- LocationCity
- AvgMonthlyLongDistanceCharges

These features exhibited the highest drift scores and statistical significance, indicating major distributional changes between training and production environments.

Observations

- Temporal drift (Month) suggests seasonality effects
- Behavioral changes (TransactionMode, DigitalInvoicing) indicate evolving user habits
- Geographic drift (LocationCity) may reflect regional shifts in customer base

Drift Table (Top Features)

Feature	Type	PSI	p_value	Stat	Severity	Drift_Detected
Contract	Categorical	25.9562	0.0	YES	🔴 HIGH	True
Month	Categorical	24.6351	0.0	YES	🔴 HIGH	True
TransactionMode	Categorical	2.7229	0.0	YES	🔴 HIGH	True
DigitalInvoicing	Categorical	2.2033	0.0	YES	🔴 HIGH	True
LocationCity	Categorical	0.4685	0.0	YES	🔴 HIGH	True
AvgMonthlyLongDistanceCharges	Numerical	0.2775	0.0	YES	🔴 HIGH	True
NumberofReferrals	Numerical	0.172	0.0	YES	🟡 MODERATE	True
ReferredaFriend	Categorical	0.105	0.0	YES	🟡 MODERATE	True
MonthlyCharge	Numerical	0.0949	0.0	YES	🟢 LOW	True
PrioritySupport	Categorical	0.0858	0.0	YES	🟢 LOW	True
TotalLongDistanceCharges	Numerical	0.0501	0.0	YES	🟢 LOW	True
TotalCharges	Numerical	0.0057	0.0	YES	🟢 LOW	True
Longitude	Numerical	0.0015	0.0114	YES	🟢 LOW	True
UserAge	Numerical	0.0014	0.8878	NO	🟢 LOW	False
DataUsageAvg	Numerical	0.0011	0.8477	NO	🟢 LOW	False

These insights highlight that drift is not random but driven by **real-world behavioral and temporal changes**, reinforcing the importance of continuous monitoring.

16. Robustness Analysis

The system demonstrates strong robustness through several design choices:

- **Multi-signal drift detection** (distribution + statistical tests) reduces false positives
- **Feature-level mitigation** ensures targeted corrections instead of global overfitting
- **A sample weighting strategy** is implemented to adapt model learning to shifted distributions.
- **Category alignment safeguards** prevent inference-time failures

Even under moderate drift conditions, the system maintains stable performance and improves generalization, indicating resilience to distributional changes.

17. System Limitations

Despite strong performance, several limitations remain:

- Extremely high drift (e.g., complete distribution shifts) may require full retraining rather than mitigation
- Correlated features are not explicitly handled, which may lead to redundant adjustments
- Drift thresholds (e.g., PSI cutoffs) are static and may not generalize across domains
- Sample weighting assumes independence between features, which may not always hold

These limitations highlight opportunities for further refinement in real-world deployments.

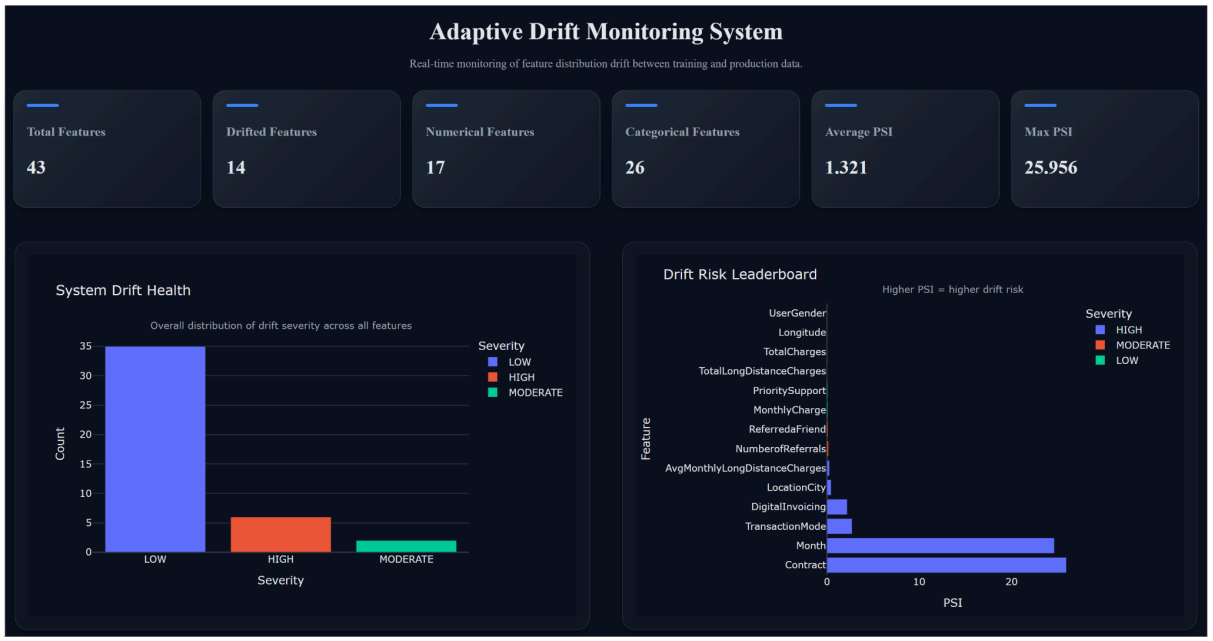
18. Future Improvements

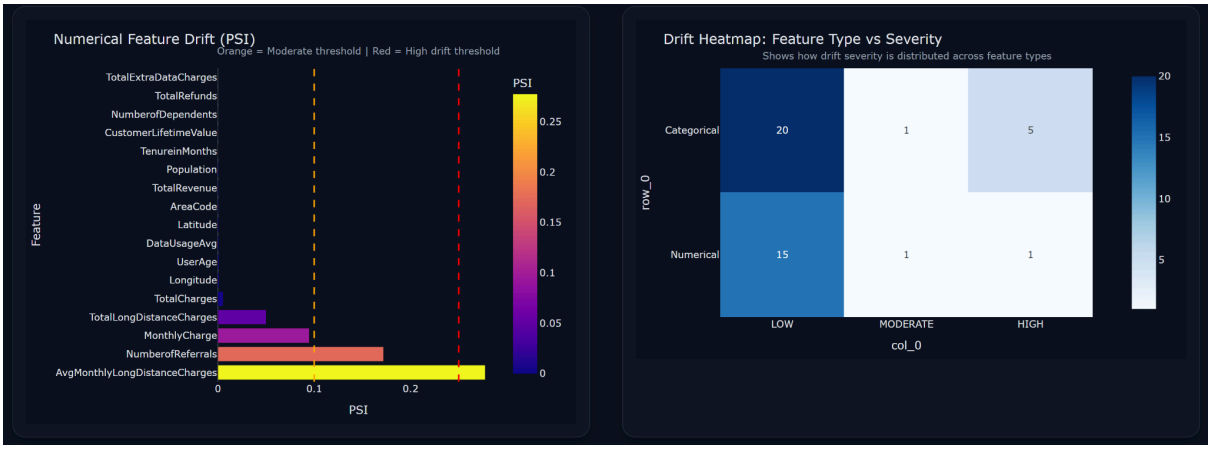
Several enhancements can further improve the system:

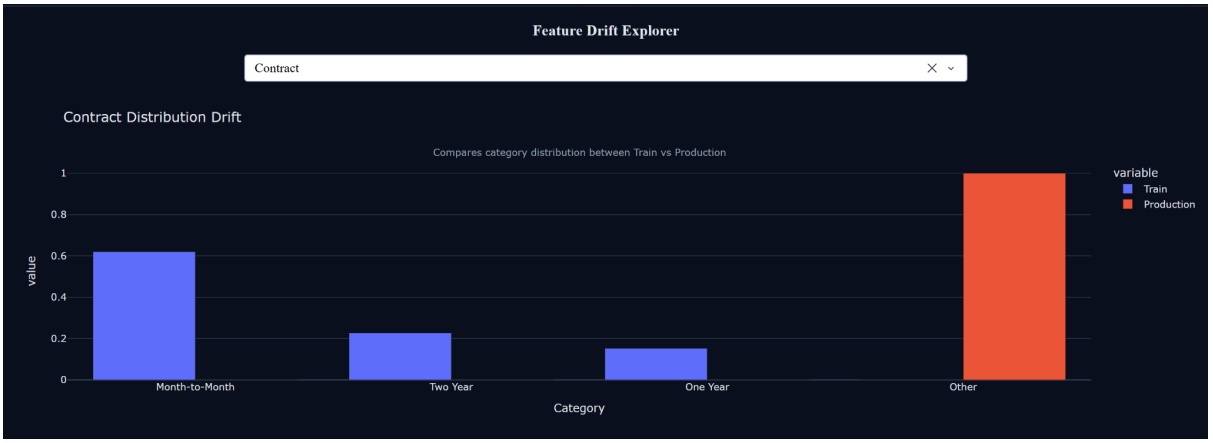
- **Adaptive drift thresholds** based on historical trends
- **Online learning models** for continuous updates
- **Advanced drift metrics** (e.g., KL divergence, MMD)
- **Feature interaction-aware mitigation**
- **Domain adaptation techniques** for better cross-distribution learning

These improvements would enhance scalability, adaptability, and performance in dynamic production environments.

19. Dashboard







19.1 Overview

An interactive dashboard was developed using Dash and Plotly to provide a clear and interpretable view of dataset drift. It serves as a centralized interface for monitoring system health, analyzing feature-level drift, and supporting data-driven decision-making.

19.2 Dashboard Components

The dashboard consists of several interactive visual components designed to provide both high-level monitoring and detailed feature-level analysis of drift.

1. KPI Summary Cards

- Provide a quick snapshot of overall system status
    - Total Features → Total number of monitored features (43)
    - Drifted Features → Number of features flagged with drift (14)
    - Numerical vs Categorical → Breakdown of feature types
    - Average PSI → Overall dataset drift level (0.123, indicating moderate drift)
    - Max PSI → Highest drift observed (notably in features such as Contract and Month)
- **Purpose**
    - Enables rapid assessment of system health without inspecting detailed charts

2. System Drift Health (Severity Distribution Chart)

- Bar chart showing counts of LOW, MODERATE, and HIGH drift features
- **Interpretation**
    - A majority of LOW drift features indicates general stability
    - Presence of HIGH drift features highlights critical areas requiring attention
    - Distribution across severity levels reflects overall system drift condition
- **Purpose**
    - Provides an immediate understanding of drift intensity across the system

3. Drift Risk Leaderboard

- Horizontal bar chart ranking features by PSI (highest at top)
- Color-coded by severity (LOW / MODERATE / HIGH)
- **Interpretation**
    - Top-ranked features (e.g., Contract, Month) are the primary contributors to dataset drift
    - Lower-ranked features indicate relatively stable distributions
- **Purpose**
    - Helps prioritize features for investigation and mitigation

4. Numerical Drift Plot

- Displays PSI values for numerical features only
- Includes threshold reference lines

    - 0.1 → Moderate drift
    - 0.25 → High drift
- **Interpretation**
    - Features exceeding 0.25 exhibit significant distribution shift
    - Features near thresholds indicate emerging instability
- **Purpose**
    - Provides focused analysis of continuous feature drift

5. Drift Heatmap (Feature Type vs Severity)

- Matrix showing count of features by
    - Rows → Feature type (Numerical / Categorical)
    - Columns → Severity (LOW / MODERATE / HIGH)
- **Interpretation**
    - Highlights whether drift is concentrated in categorical or numerical features
    - Example insights:
        - High categorical drift → possible encoding or category distribution changes
        - High numerical drift → distribution or scaling shifts
- **Purpose**
    - Provides structural insight into drift patterns across feature groups

6. Feature Drift Explorer (Interactive Component)

- Dropdown allows selection of any drifted feature
- Visualization updates dynamically based on feature type
- **For Categorical Features**
    - Grouped bar chart comparing Train vs Production distributions
        - X-axis → Categories
        - Y-axis → Proportion (normalized counts)
    - **Interpretation**
        - Differences in category proportions indicate distribution shift
        - Missing or new categories may signal upstream data issues
- **For Numerical Features**
    - Overlaid histogram comparing Train vs Production distributions
    - **Interpretation**
        - Shift in mean → horizontal displacement
        - Change in spread → variance shift
        - Shape differences → distributional changes (e.g., skewness)
- **Purpose**
    - Enables detailed investigation of individual feature drift
    - Bridges the gap between statistical detection and visual explanation

19.3 Design and Accessibility

The dashboard adopts a modern dark-themed interface with card-based layouts, consistent styling, and annotated charts to enhance readability. The layout is responsive and structured to support efficient analysis.

In addition to the live interactive dashboard, a static HTML version is generated, allowing results to be easily shared and reviewed offline.

19.4 Summary

Overall, the dashboard transforms raw drift metrics into an intuitive and visually rich monitoring tool, improving interpretability and enabling effective analysis of data drift in real-world scenarios.

20. Conclusion

This project presents a comprehensive system for **detecting, analyzing, and mitigating data drift** in telecom churn prediction.

The proposed pipeline integrates:

- Multi-method drift detection
- Feature-level and global mitigation strategies
- LightGBM-based predictive modeling
- Quantitative performance validation

The system successfully improved test AU-PRC by **+0.0160**, demonstrating the effectiveness of the drift-aware approach.

Overall, the solution provides a **robust, scalable, and production-ready framework** for maintaining model performance under changing data distributions.